Degree lower bound for polynomials computing square roots modulo primes congruent to 1 mod 4

Kiran Kedlaya, Swastik Kopparty

doi:10.4230/LIPIcs.CCC.2024.25; CC BY 4.0

Allowed standard prerequisites (editorial).

- Polynomial vanishing sets over finite fields and formal differentiation. Author Sections 1-3; original full bibliography retained

Theorem 1.1. *Let $p \equiv 1 \mod 4$. Then any polynomial that computes square roots in $\mathbb{F}_p$ has degree at least $\frac{p-1}{3}$.*

Proof. Suppose $f(X)$ is of degree $d < \frac{p-1}{3}$ and computes square roots in $\mathbb{F}_p$. Then, since $X^{(p-1)/2}-1$ is the vanishing polynomial of the set of nonzero perfect squares in $\mathbb{F}_p$, we have:

$$f(X)^2 - X \equiv 0 \mod (X^{(p-1)/2} - 1).$$

Let $A(X)$ be the polynomial of degree $2d - (p-1)/2$ such that

$$f(X)^2 - X = A(X) \cdot (X^{(p-1)/2} - 1).$$

Let $B(X) = X - A(X)$. Then we get:

$$f(X)^2 = A(X) \cdot X^{(p-1)/2} + B(X), \tag{1}$$

where:

- $\deg(f(X)) = d$.
- $\deg(A(X)), \deg(B(X)) \leq 2d - (p-1)/2$.
- $A(X) \neq 0$. If $A(X) = 0$ then $f(X)^2 = X$, which is impossible for a polynomial $f(X)$.
- $B(X) \neq 0$. Otherwise $A(X) = X$, and $f(X)^2 = X^{(p+1)/2}$, which is possible only if $p \equiv 3 \mod 4$.

These conditions together will give us our lower bound on d.
Taking derivatives[1] of both sides of (1), we get:

$$2f(X)f'(X) = A'(X) \cdot X^{(p-1)/2} - \frac{1}{2}A(X)X^{(p-3)/2} + B'(X). \tag{2}$$

[1] Throughout this paper, we work with formal derivatives of polynomials.

Computing $2f(X)^2 f'(X)$ in two ways using (1) and (2), we get:

$$2f'(X)A(X)X^{(p-1)/2}+2f'(X)B(X) = f(X)\left(X\cdot A'(X) - \frac{1}{2}A(X)\right)X^{(p-3)/2}+f(X)B'(X).$$

Now, using our assumption on d, the degrees of $2f'(X)B(X)$ and $f(X)B'(X)$ are both at most $3d-(p-1)/2-1 < (p-3)/2$, and thus taking the above equation mod $X^{(p-3)/2}$,

$$2f'(X)B(X) = f(X)B'(X).$$

Since $B(X) \neq 0$, we get $\frac{2f'(X)}{f(X)} = \frac{B'(X)}{B(X)}$. Since $2\deg(f), \deg(B) < p$, by a basic property of logarithmic derivatives, this implies $f(X)^2 = \lambda B(X)$ for some nonzero λ, contradicting the fact that $A(X) \neq 0$. (See Remark 1 in Section for a precise statement and a proof.)

Thus our assumption that $d < \frac{p-1}{3}$ is wrong, and the theorem follows. □

Let p be an odd prime, and let $\mathbb{F}_p$ be the finite field with p elements.

We say $f(X) \in \mathbb{F}_p[X]$ *computes square roots* in $\mathbb{F}_p$, if for all nonzero perfect squares $a \in \mathbb{F}_p$, we have:

$$f(a)^2 = a.$$

In other words, for each nonzero perfect square $a \in \mathbb{F}_p$, $f(a)$ is one of its two square roots.

Logarithmic derivatives

The key fact about logarithmic derivatives that we are using is that if $f(X), B(X) \in \mathbb{F}_p[X]$, $t\cdot\deg(f), \deg(B) < p$, and:

$$t\frac{f'(X)}{f(X)} = \frac{B'(X)}{B(X)},$$

then $f(X)^t = \lambda \cdot B(X)$ for some constant $\lambda \in \mathbb{F}_p$.

We recap a quick proof. The hypothesis implies that $\left(\frac{f(X)^t}{B(X)}\right)' = 0$, and thus $\frac{f(X)^t}{B(X)}$ must be a rational function in X^p. To see the last deduction, note that $\left(f(X)^t\cdot B(X)^{p-1}\right)' = \left(\frac{f(X)^t}{B(X)}\cdot B(X)^p\right)' = \left(\frac{f(X)^t}{B(X)}\right)'\cdot B(X)^p + \frac{f(X)^t}{B(X)}\cdot p\cdot B(X)^{p-1}\cdot B'(X) = 0+0 = 0$. Thus the polynomial $f(X)^t\cdot B(X)^{p-1}$ is a polynomial in X^p, which implies that $\frac{f(X)^t}{B(X)} = \frac{f(X)^t\cdot B(X)^{p-1}}{B(X)^p} = \frac{f(X)^t\cdot B(X)^{p-1}}{B(X^p)}$ is a rational function in X^p. Once we know that $\frac{f(X)^t}{B(X)}$ is a rational function in X^p, our assumption about the degrees implies the result.

References

[ADN03] Simon Joseph Agou, Marc Deléglise, and Jean-Louis Nicolas. Short polynomial representations for square roots modulo p. *Designs, Codes and Cryptography*, 28(1):33–44, 2003.

[AS97] Sanjeev Arora and Madhu Sudan. Improved low-degree testing and its applications. In *Proceedings of the twenty-ninth annual ACM symposium on Theory of computing*, pages 485–495, 1997.

[Bir00] András Biró. On polynomials over prime fields taking only two values on the multiplicative group. *Finite Fields and Their Applications*, 6(4):302–308, 2000.

[BS96] Eric Bach and Jeffrey Outlaw Shallit. *Algorithmic number theory: Efficient algorithms*, volume 1. MIT press, 1996.

[Car11] NA Carella. Formulas for the square root modulo p. *arXiv preprint arXiv:1101.4605*, 2011.

[CKL15] Seunghwan Chang, Bihtnara Kim, and Hyang-Sook Lee. Polynomial representations for n-th roots in finite fields. *Journal of the Korean Mathematical Society*, 52(1):209–224, 2015.

[CS00] Don Coppersmith and Igor Shparlinski. On polynomial approximation of the discrete logarithm and the diffie—hellman mapping. *Journal of Cryptology*, 13:339–360, 2000.

[GR05] Venkatesan Guruswami and Atri Rudra. Limits to list decoding reed-solomon codes. In *Proceedings of the thirty-seventh annual ACM symposium on Theory of computing*, pages 602–609, 2005.

[GS98] Venkatesan Guruswami and Madhu Sudan. Improved decoding of reed-solomon and algebraic-geometric codes. In *Proceedings 39th Annual Symposium on Foundations of Computer Science (Cat. No. 98CB36280)*, pages 28–37. IEEE, 1998.

[Kal91] Erich Kaltofen. Effective noether irreducibility forms and applications. In *Proceedings of the twenty-third annual ACM symposium on Theory of Computing*, pages 54–63, 1991.

[Mas84] Richard Clive Mason. *Diophantine equations over function fields*, volume 96. Cambridge University Press, 1984.

[MW86] Gary Mullen and David White. A polynomial representation for logarithms in gf (q). *Acta arithmetica*, 3(47):255–261, 1986.

[Sch06] Wolfgang M Schmidt. *Equations over finite fields: an elementary approach*, volume 536. Springer, 2006.

[Sha73] Daniel Shanks. Five number-theoretic algorithms. In *Proceedings of the Second Manitoba Conference on Numerical Mathematics (Winnipeg), 1973*, 1973.

[Sho09] Victor Shoup. *A computational introduction to number theory and algebra*. Cambridge university press, 2009.

[Sto81] W Wilson Stothers. Polynomial identities and hauptmoduln. *The Quarterly Journal of Mathematics*, 32(3):349–370, 1981.

[Ton91] Alberto Tonelli. Bemerkung über die auflösung quadratischer congruenzen. *Nachrichten von der Königl. Gesellschaft der Wissenschaften und der Georg-Augusts-Universität zu Göttingen*, 1891:344–346, 1891.

[VZGG13] Joachim Von Zur Gathen and Jürgen Gerhard. *Modern computer algebra*. Cambridge university press, 2013.

[WB83] Lloyd R Welch and Elwyn R Berlekamp. Error correction for algebraic block codes. *US patent*, (4,633,470), 1983.

[Win02] Arne Winterhof. Polynomial interpolation of the discrete logarithm. *Designs, Codes and Cryptography*, 25(1):63–72, 2002.